Learning Objectives

- How should managers build a business case for the acquisition and development of new information systems?
- What are the objectives of project management, and why is it so essential in developing information systems?
- What are the principal risk factors in information systems projects?
- How can project risk be managed?
- How will MIS help my career?

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Students – please read this Case Study – next 2 slides - (take 5 minutes)

BDO Canada Selects New Project Management Software

BDO Canada LLP is part of the BDO global network of public accounting, tax, consulting, and business advisory firms. It is a leading Canadian accounting and advisory firm, serving a broad range of clients across the country. There are nearly 100 people in its IT department, and members are engaged in numerous projects.

Resource management has historically been one of the IT department's greatest challenges. BDO had been using an internally built project management tool for five years to ensure its projects were tracked accurately and completely, that resources were properly allocated, and that the IT department could plan and forecast better. In 2016, BDO's Project Management Office (PMO) team determined that the homegrown tool lacked this functionality and no longer followed best practices. Most team members used different project scheduling and tracking methods, which prevented the PMO team from having an overall view of projects. A user-friendly online software-as-a-service (SaaS) tool would work better with BDO's growing portfolio of cloud applications.

BDO's IT department considered a number of different solutions, using external consultants as well as its own internal staff. They issued a request-for-proposal for five project management products. After seeing product demos, sandbox testing (which eliminates risks during testing by isolating the test environment from the production environment), and meeting with project management software vendors, the BDO PMO team selected Microsoft Project Online.

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Project Online works seamlessly with other tools in the Microsoft Office 365 SaaS suite. For example, BDO's IT staff can create a project in Project Online and trigger Microsoft Flow to automatically create a topic in Microsoft Teams, the hub for teamwork in Office 365. Employees having conversations in Microsoft Teams are able to access information directly from Microsoft SharePoint Online and obtain project and portfolio reporting from Microsoft Power BI. SharePoint Online is a cloud-based service, hosted by Microsoft, enabling users to create sites to share documents and information with colleagues, partners, and customers. Microsoft Power BI is a business analytics software service providing interactive data visualization and business intelligence capabilities with a user-friendly interface. Project Online enables BDO's CIO Kim Watt and others to have a clear view of projects that are delayed or on time, centralized risk issues, and other status updates.

The PMO team rolled out Project Online to the BDO IT department in early 2019. To promote adoption, BDO's senior IT management meets on a regular basis, and the PMO team works with IT project managers to help them with product features and processes. The PMO team also offers drop-in sessions for teams with any questions two or three times per month.

After implementing Microsoft Project Online, BDO was quick to see benefits. With just a few steps, CIO Watt and her colleagues can run a resource report to see which teams are busy and who has time to assist with existing projects or initiate new ones. The PMO team, which manages upcoming project tasks and resources, can quickly refer to an up-to-date schedule maintained in a centralized location. BDO's IT Team can run cost-benefit analyses early in a project's life cycle to make sure it's worthwhile before the project team invests too much time on it. Project Online informs discussions about the best way to prioritize limited resources. Projects can be prioritized before they are started based on the needs of the business.

Sources: "BDO," customers.microsoft.com, accessed May 6, 2020, and bdo.ca,

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BDO Canada Selects New Software

- **Problem**
  - ✓ Need for enterprise-wide resource management
  - ✓ Outdated homegrown project management software
  - ✓ Multiple project scheduling and tracking methods
- **Solutions**
  - ✓ Track projects
  - ✓ Allocate resources
  - ✓ Assemble project team: internal staff and external IT consultants
  - ✓ Adopt Microsoft Project Online, integrated with other Microsoft tools: Microsoft Teams, Microsoft SharePoint, Microsoft Power BI
- **Online Project Management System**
- **Illustrates special managerial and organizational techniques required to make projects to build or improve information systems effective**

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Building a Business Case for the Acquisition and Development of a New Information System

- **Important Elements of a Business Case**
  - ✓ Describes problem facing the organization
  - ✓ Provides analysis of costs, benefits, and risks
  - ✓ Describes rationale for proceeding with investment
  - ✓ Shows how investment supports firm's strategic goals
  - ✓ Provides information necessary to make informed decision
  - ✓ Explains how investment will provide value for business
  - ✓ Identifies risks and alternative solutions
  - ✓ Describes how solution may require changes in organizational culture, systems, processes, jobs

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Factors to Consider in Making the Business Case

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Information Systems Plan

- **Identifies systems projects that will deliver most business value**
- **Links development to business plan**
- **Road map indicating direction of systems development, includes:**
  - ✓ Purpose of plan
  - ✓ Strategic business plan rationale
  - ✓ Current systems/situation
  - ✓ New developments
  - ✓ Management strategy
  - ✓ Implementation plan
  - ✓ Budget
- **For effective plan**
  - ✓ Document existing systems and infrastructure components
  - ✓ Identify decision-making improvements
  - ✓ Develop metrics established for quantifying values
  - ✓ Clear understanding of long-term and short-term information requirements
  - ✓ Describe organizational changes required

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Portfolio Analysis

- **Used to evaluate alternative system projects (BUILD vs BUY)**
- **Inventories all of the organization's information systems projects and assets**
- **Each system has profile of risk and benefit**
  - ✓ High benefit, low risk
  - ✓ High benefit, high risk
  - ✓ Low benefit, low risk
  - ✓ Low benefit, high risk
- **To improve return on portfolio, balance risk and return from systems investments**

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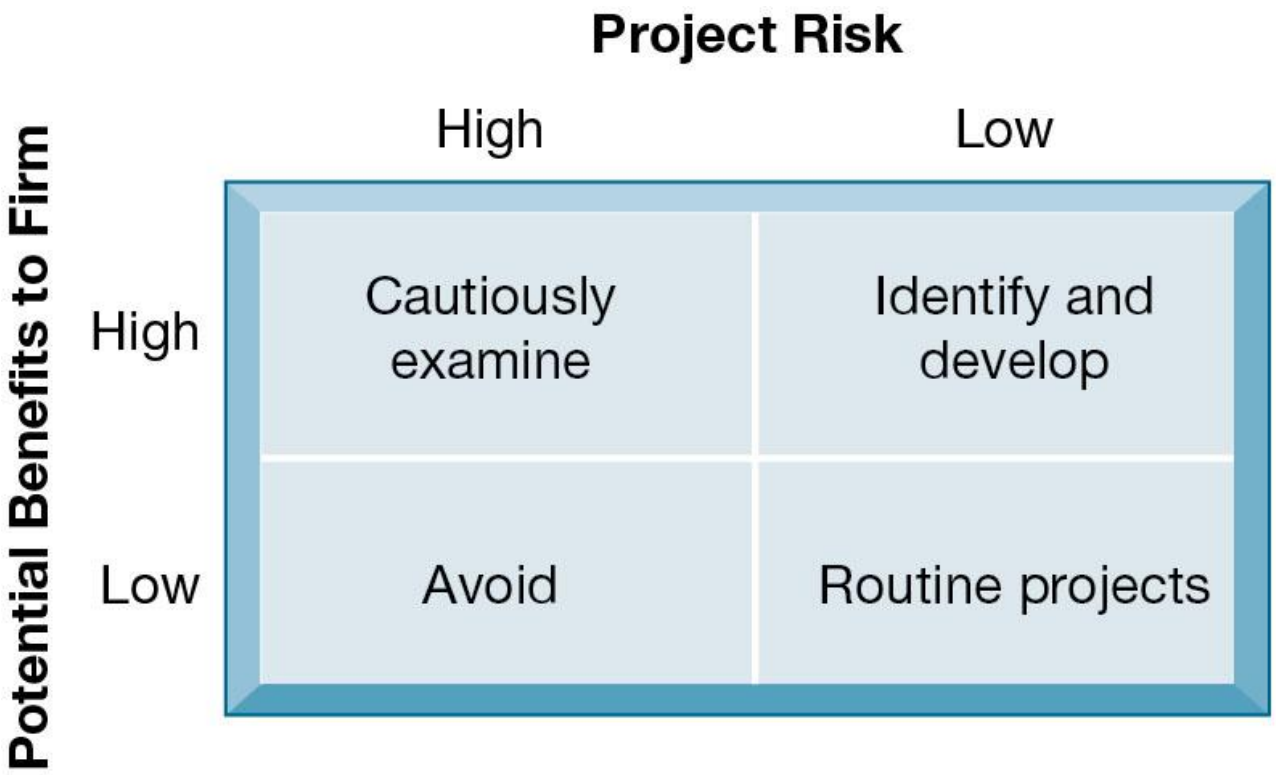
A System Portfolio



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Scoring Models

- Used to evaluate alternative system projects, especially when many criteria exist
- Assigns weights to various features of system and calculates weighted totals
- Many qualitative judgments involved
- Requires experts who understand the issues and the technology

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Example of a Scoring Model for an ERP System

Criteria	Weight	E R P System A %	E R P System A Score	E R P System B %	E R P System B Score
1.1 Online order entry	4	67	268	73	292
1.2 Online pricing	4	81	324	87	348
1.3 Inventory check	4	72	288	81	324
1.4 Customer credit check	3	66	198	59	177
1.5 Invoicing	4	73	292	82	328
2.1 Production forecasting	3	72	216	76	228
2.2 Production planning	4	79	316	81	324
(etc.)	(etc.)	(etc.)	(etc.)	(etc.)	(etc.)
Grand Totals			3,128		3,300

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Determining Solution Costs and Benefits

- **Information System Costs and Benefits**
  - ✓ ***Tangible Benefits***
    - Can be quantified and assigned monetary value
    - Typically produced by systems that displace labor, save space
      - E.g. transaction and clerical systems
  - ✓ ***Intangible Benefits***
    - Cannot be immediately quantified but may lead to quantifiable gains in the long run
      - Example: more efficient customer service, enhanced decision making
    - Typically produced by systems that influence decision making
      - E.g. ESS, DSS, collaborative work systems

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Capital Budgeting for Information Systems

- **Capital Budgeting Models**
  - ✓ Measure value of long-term capital investment projects
  - ✓ Rely on measures of the firm's cash outflows and inflows
- **Principal Capital Budgeting Models Used to Evaluate IT Projects**
  - ✓ Payback method
  - ✓ Accounting rate of return on investment (ROI)
  - ✓ Net present value
  - ✓ Internal rate of return (IRR)
- **Limitations of Financial Models**

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Runaway Projects and System Failure

- **Runaway Projects**
  - ✓ Exceed budget, schedule
    - McKinsey/Oxford University study: ran 66 percent over budget; 33 percent over schedule
- **Types of System Failure**
  - ✓ Fail to capture essential business requirements
  - ✓ Fail to provide organizational benefits
  - ✓ Complicated, poorly organized user interface
  - ✓ Inaccurate or inconsistent data

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Consequences of Poor Project Management

Cost overruns
Time slippage
Technical shortfalls impairing performance
Failure to obtain anticipated benefits

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Project Management Objectives

- **Project**
  - ✓ planned series of related activities for achieving a specified business objective
- **Project Management**
  - ✓ Activities include planning work, assessing risk, estimating resources required, organizing the work, assigning tasks, controlling project execution, reporting progress, analyzing results
- **Five Major Variables**
  - ✓ Scope
  - ✓ Time
  - ✓ Cost
  - ✓ Quality
  - ✓ Risk

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Dimensions of Project Risk

- **Project Size**
  - ✓ Cost
  - ✓ Time
  - ✓ Number of organizational units affected
  - ✓ Organizational complexity
- **Project Structure**
  - ✓ Structured, defined requirements run lower risk
- **Experience with Technology**
  - ✓ Team familiar with hardware and software

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Change Management and the Concept of Implementation

- **Change Management**
  - ✓ Required for successful system building
  - ✓ New information systems have powerful behavioral and organizational impact
- **Implementation**
  - ✓ All organizational activities working toward adoption, management, and routinization of an innovation
- **Change Agent**
  - ✓ One role of systems analyst
  - ✓ Redefines the configurations, interactions, job activities, and power relationships of organizational groups

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Change Management and the Concept of Implementation (continued)

- **Role of End Users**
  - ✓ With high levels of user involvement
    - System more likely to conform to requirements
    - Users more likely to accept system
- **User–Designer Communication Gap**
  - ✓ Users and information systems specialists
- **Management (Stakeholders) Support and Commitment**
  - ✓ Affects positive perception by both users and technical staff
  - ✓ Ensures sufficient funding and resources
  - ✓ Helps enforce required organizational changes

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Change Management and the Concept of Implementation (continued)

- **Change management challenges for business process management (BPM), enterprise applications, and mergers and acquisitions**
  - ✓ *Very high failure rate among enterprise applications and BPM projects (up to 70 percent for far-reaching BPM)*
    - Poor implementation and change management practices
      - Employee concerns about change
      - Resistance by key managers
      - Changing job functions, career paths, recruitment practices
  - ✓ *Mergers and Acquisitions*
    - Similarly high failure rate of integration projects
    - Merging of systems of two companies requires:
      - Considerable organizational change, complex systems projects

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Managing Project Risks

- **First step involves identifying nature and level of risk of project**
- **Each project can then be managed with tools and risk-management approaches geared to level of risk**
- **Managing Technical Complexity**
  - ✓ Internal integration tools
    - Project leaders with technical and administrative experience
    - Highly experienced team members
    - Frequent team meetings
    - Securing of technical experience outside firm if necessary

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Formal Planning and Control Tools

- **Used for documenting and monitoring project plans**
- **Help identify bottlenecks and impact of problems**
- **Gantt Charts**
  - ✓ Visual representation of timing and duration of tasks
  - ✓ Human resource requirements of tasks
- **PERT Charts**
  - ✓ Graphically depict tasks and interrelationships
  - ✓ Indicate sequence of tasks necessary

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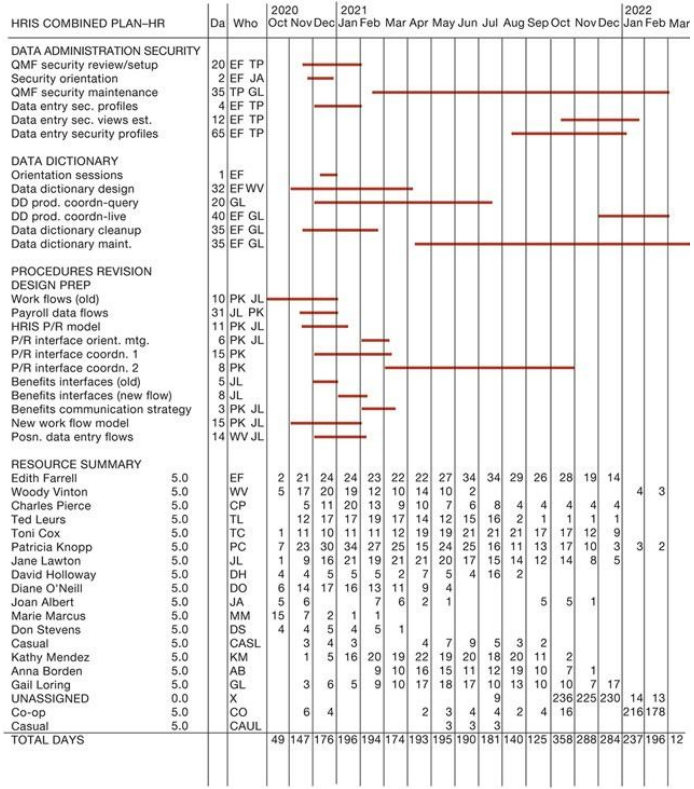
A Gantt Chart



HRIS COMBINED PLAN–HR		Da	Who	2020 Oct	Nov	Dec	2021 Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	2022 Jan	Feb	Mar
DATA ADMINISTRATION SECURITY																					
QMF security review/setup		20	EF TP																		
Security orientation		2	EF JA																		
QMF security maintenance		35	TP GL																		
Data entry sec. profiles		4	EF TP																		
Data entry sec. views est.		12	EF TP																		
Data entry security profiles		65	EF TP																		
DATA DICTIONARY																					
Orientation sessions		1	EF																		
Data dictionary design		32	EF WV																		
DD prod. coordn-query		20	GL																		
DD prod. coordn-live		40	EF GL																		
Data dictionary cleanup		35	EF GL																		
Data dictionary maint.		35	EF GL																		
PROCEDURES REVISION																					
DESIGN PREP																					
Work flows (old)		10	PK JL																		
Payroll data flows		31	JL PK																		
HRIS P/R model		11	PK JL																		
P/R interface orient. mtg.		6	PK JL																		
P/R interface coordn. 1		15	PK																		
P/R interface coordn. 2		8	PK																		
Benefits interfaces (old)		5	JL																		
Benefits interfaces (new flow)		8	JL																		
Benefits communication strategy		3	PK JL																		
New work flow model		15	PK JL																		
Posn. data entry flows		14	WV JL																		
RESOURCE SUMMARY																					
Edith Farrell	5.0		EF	2	21	24	24	23	22	22	27	34	34	29	26	28	19	14			
Woody Vinton	5.0		WV	5	17	20	19	12	10	14	10	2							4	3	
Charles Pierce	5.0		CP		5	11	20	13	9	10	7	6	8	4	4	4	4	4			
Ted Leurs	5.0		TL		12	17	17	19	17	14	12	15	16	2	1	1	1	1			
Toni Cox	5.0		TC	1	11	10	11	11	12	19	19	21	21	21	17	17	12	9			
Patricia Knopp	5.0		PC	7	23	30	34	27	25	15	24	25	16	11	13	17	10	3	3	2	
Jane Lawton	5.0		JL	1	9	16	21	19	21	21	20	17	15	14	12	14	8	5			
David Holloway	5.0		DH	4	4	5	5	5	2	7	5	4	16	2							
Diane O'Neill	5.0		DO	6	14	17	16	13	11	9	4										
Joan Albert	5.0		JA	5	6			7	6	2	1				5	5	1				
Marie Marcus	5.0		MM	15	7	2	1	1													
Don Stevens	5.0		DS	4	4	5	4	5	1												
Casual	5.0		CASL		3	4	3			4	7	9	5	3	2						
Kathy Mendez	5.0		KM		1	5	16	20	19	22	19	20	18	20	11	2					
Anna Borden	5.0		AB					9	10	16	15	11	12	19	10	7	1				
Gail Loring	5.0		GL		3	6	5	9	10	17	18	17	10	13	10	10	7	17			
UNASSIGNED	0.0		X										9			236	225	230	14	13	
Co-op	5.0		CO		6	4				2	3	4	4	2	4	16			216	178	
Casual	5.0		CAUL								3	3	3								
TOTAL DAYS				49	147	176	196	194	174	193	195	190	181	140	125	358	288	284	237	196	12

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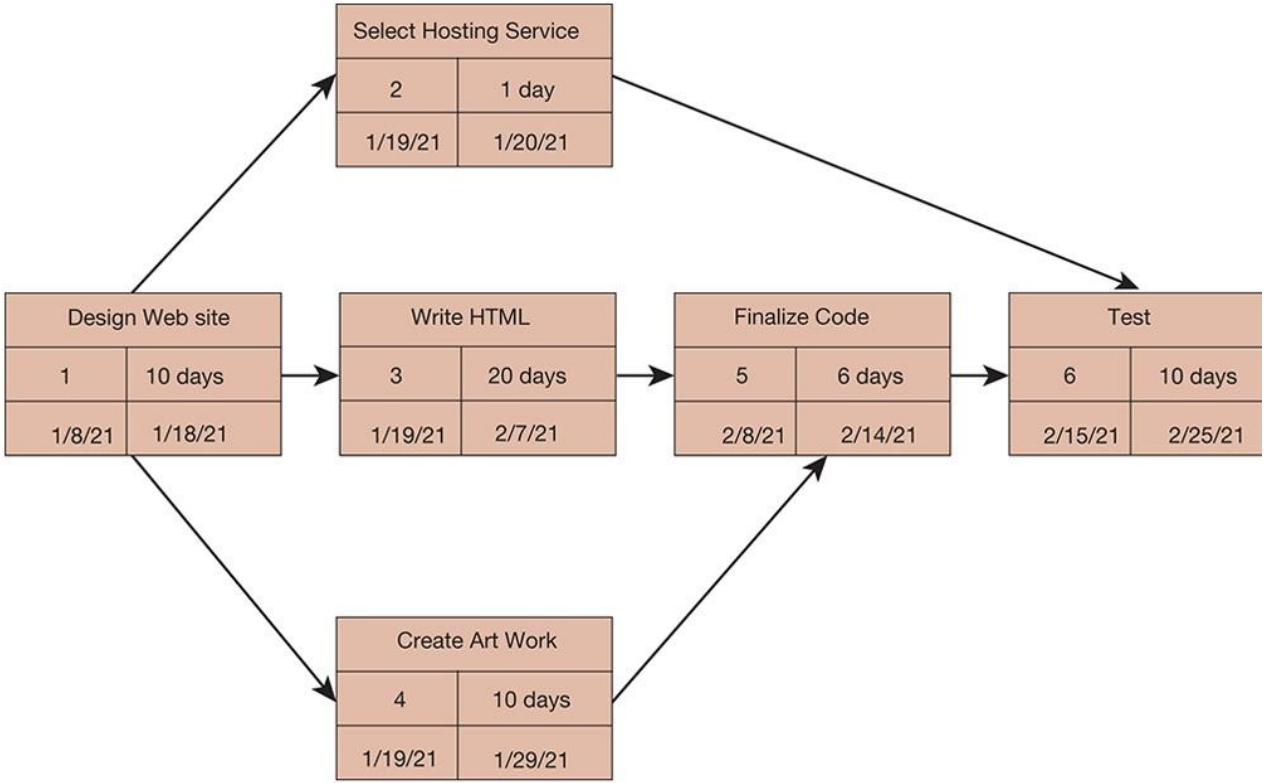
A PERT Chart



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Increasing User Involvement and Overcoming User Resistance

- **External Integration Tools**
  - ✓ Link work of implementation team to users at all levels
- **User Resistance to Organizational Change|**
  - ✓ Counterimplementation
- **Strategies to Overcome User Resistance**
  - ✓ User participation, education and training
  - ✓ Management edicts and policies
  - ✓ Incentives for cooperation
  - ✓ Improvement of end-user interface
  - ✓ Resolution of organizational problems prior to introduction of new system

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Designing for the Organization

- **Need to Address:**
  - ✓ Ways in which organization changes with new system
- **Ergonomics**
  - ✓ Interaction of people and machines/BPM in work environment
- **Organizational Impact Analysis**
  - ✓ How system will affect organizational structure, attitudes, decision making, operations
- **Sociotechnical Design**
  - ✓ Addresses human and organizational issues

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Students – please read this Case Study – next 3 slides - (take 5 minutes)

Interactive Session Management

Sauder Woodworking Gets ERP Implementation Right

Sauder Woodworking is North America's leading producer of ready-to-assemble furniture, and one of the top five U.S. residential furniture manufacturers. 90 percent of Sauder furniture is manufactured in Archbold, Ohio, where the company was founded in 1934. (Sauder also sources furniture from a network of global suppliers.) Its Ohio facility has 4-millionsquare- foot floor space housing high-tech furnituremaking equipment from around the world and 2,000 employees. The company offers nearly 50 furniture collections. Furniture-making requires skilled design, artistry, and attention to detail, and Sauder has historically excelled in these areas.

Sauder started using SAP ERP Central Component (ECC) software for enterprise resource planning (ERP) in 2004. ECC provides modules covering a full range of industry applications, including finance, logistics, HR, product planning, and customer service, linked together into a single, customizable system run on a database of the user's choice. As a modular system, SAP ECC is designed so that organizations can use the pieces they need, configured in a way that makes sense for their business. Sauder used a phased implementation approach, starting with the modules for order-to-cash (business processes for receiving and processing customer orders). Sauder finished implementing all the main SAP ECC modules by 2015.

By that time, SAP had released SAP S/4HANA, a more leading-edge version of its ERP system that features in-memory computing (see Chapter 6). Sauder management had to decide whether to optimize the SAP modules it had already implemented on premises or whether to switch to S/4HANA to take advantage of its new functionality. Switching to the newest version of SAP ERP software would require more work than sticking with Sauder's existing system. Management opted for an on-premises version of SAP S/4HANA so that it could implement the new S/4HANA suite on its own timeline rather than wait and then be pressured to change software when SAP withdrew support

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for the older ECC system.

Another decision was whether to go greenfield (a completely new implementation) or brownfield (converting the existing system to the new one). A greenfield implementation would require installing a clean installation of the ERP system, then importing all the relevant data. A brownfield implementation would entail converting the existing system to the new one, using the data from the ECC system. Sauder opted for the brownfield approach. A key factor in the decision to go brownfield was the number of employees (2,000) who would be impacted by having to deal with an entirely new system. They might have trouble learning the system or resist working with the system because it was so unfamiliar. Sauder didn't want to have to deal with a lot of change management.

Sauder was quite familiar with SAP software, which was the company's business processing engine. Sauder's SAP system supported over 1,600 users conducting business transactions with the software, including those for finance, supply chain management, and shop floor interactions with SAP. In 2018, Sauder's SAP system processed 1,624,169 shipments, 1,661,225 sales orders, 14,438,089 warehouse moves, 894,664 production receipts, and 1,624,684 invoices.

There were challenges: To minimize business disruptions, management wanted the SAP ECC system converted to SAP S/4HANA within 72 hours, a very narrow time frame for a project of this scope. There was no room for error. At the same time Sauder began transitioning to SAP S/4HANA, it also made major changes to its IT infrastructure. The company migrated off legacy IBM iSeries systems to a new platform based on Dell servers and storage area networks (SANs) in a VMware virtual environment (see **Chapter 5**). A SAN is a network of storage devices that can be accessed by multiple servers or computers, providing a shared pool of storage space. Sauder's IT staff had to learn how to support these new technologies, but was fortunately assisted by Symmetry consultants. Symmetry helps firms manage complex

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SAP implementations on a global scale. Symmetry had provided services to Sauder for over 12 years, and understood the nature of its business and technology environment.

The system was successfully converted within the 72-hour downtime window, including conversion of the production planning and detailed scheduling functionality of the SAP® Advanced Planning and Optimization component within SAP S/4HANA. The cut over to the new S/4HANA system went very smoothly, with minimal disruption to the business, and the project was delivered on time and within budget.

The company plans to optimize its use of the S/4HANA system in phases, starting with finance functions. Running on SAP S/4HANA has positioned Sauder Woodworking better for achieving its strategic goals.

Sources: Eric Kavanah, "Measure Twice, Cut Once: Getting a Modern ERP Deployment Right," *eWeek*, February 11, 2020; "SAP S/4HANA Migration Case Study: Sauder Woodworking," symmetrycorp. com, accessed May 9, 2020; and "Sauder Woodworking," itelligence.com, accessed May 8, 2020.

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Sauder Woodworking Questions

1. **Why is an ERP system so important for Sauder Woodworking?**
2. **Why did Sauder want to switch to a newer ERP system?**
3. **Was SAP S/4HANA a good choice for Sauder? Why or why not?**
4. **Were there any risks in this project? How did Sauder deal with them?**

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Sauder Woodworking Gets ERP Implementation Right

1. Why is an ERP system so important for Sauder Woodworking?

- ✓ Sauder started using SAP ERP Central Component (ECC) software for enterprise resource planning (ERP) in 2004. ECC provides modules covering a full range of industry applications, including finance, logistics, HR, product planning, and customer service, linked together into a single, customizable system run on a database of the user's choice. As a modular system, SAP ECC is designed so that organizations can use the pieces they need, configured in a way that makes sense for their business.

2. Why did Sauder want to switch to a newer ERP system?

- ✓ They wanted to move to the newer version to leverage some of the new features such as in-memory computing.

3. Was SAP S/4HANA a good choice for Sauder? Why or why not?

- ✓ This was generally considered a good choice for Sauder, but the company plans to optimize its use of the S/4HANA system in phases, starting with the finance functions. Running on SAP S/4HANA has positioned Sauder Woodworking better for achieving its strategic goals.

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Sauder Woodworking Gets ERP Implementation Right (cont)

4. Were there any risks in this project? How did Sauder deal with them?

- ✓ To minimize business disruptions, management wanted the SAP ECC system converted to SAP S/4HANA within 72 hours, a very narrow time frame for a project of this scope. There was no room for error. At the same time Sauder began transitioning to SAP S/4HANA, it also made major changes to its IT infrastructure. The company migrated off legacy IBM iSeries systems to a new platform based on Dell servers and storage area networks (SANs) in a VMware virtual environment. Sauder's IT staff had to learn how to support these new technologies, but was fortunately assisted by Symmetry consultants. Symmetry helps firms manage complex SAP implementations on a global scale. Symmetry had provided services to Sauder for over 12 years and understood the nature of its business and technology environment.

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Project Management Software Tools

- **Can automate many aspects of project management**
- **Capabilities for defining, ordering tasks**
  - ✓ Assigning resources to tasks, tracking progress
  - ✓ Manage very large numbers of tasks and relationships
- **Microsoft Project**
- **Cloud-based Software**
- **Project Portfolio Management Software**

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High Level View of Technology Careers

Chief Information Officer (CIO)						
IT Director -or- VP of IT -or- Director - Technology Services						
Systems Engineering		**Systems Development**	**Project Management**	**Security**	**Support**	**Data Engineering**
Technology Operations		***Software***	***Company Priorities***	***"Are We Safe"?***	***Client and Employee Services***	***Information***
Hardware	**Infrastructure**					
Servers	Network	Applications Development	Planning	Company Policies	Production Support	Data Quality Assurance
Datacenter	Telecommunications	Web Development	Scheduling	Compliance	Desktop Support	Reporting
Cloud	Systems Performance	Vendor Software Specialists	Priority Management	Network Security		Database Management
			Business Analysis	Data Security		CRM Tools (ie Salesforce)
	DevOps Specialists		App Quality Assurance	Risk Management		
	Integration	Automation	Agile/Waterfall/ etc..			
	Systems Optimization	Systems Quality Assurance				
Skillset and Careers						
Systems Engineer	Network Engineer	Programmer	PMP Specialist		Microsoft Engineer	Data Scientists
Cloud Engineer	Operating Systems	Languages	Business Analyst			Report Programmers
Systems Software Specialist						
Certification Driven		**Not Certification Driven**	**PMP Certification**	**Certification Driven**	**Certification Driven**	**PhD / Mathematicians**

CIO – Strategic Role

IT Director or VP – IT …. Day to Day role

Added column – Database Administrators

KAHOOT TIME!!!

Assignment

Read the Case Study on next 4 slides and answer the three(3) questions that follow.

Arup Moves Project Management to the Cloud

Arup Group Limited is a multinational professional services firm headquartered in London that provides engineering, design, planning, project management, and consulting services for all aspects of structures and environments of human construction. Founded in 1946, Arup now has over 16,000 staff based in 96 offices across 40 countries throughout the globe. The company defines itself as one where professionals of diverse disciplines—engineers, planners, designers, financial experts, consulting specialists, and sustainability professionals—can work together to deliver projects and services of greater quality than by working in isolation. Arup has worked on projects in over 160 countries, including the Pompidou Center in Paris, the Sydney Opera House, and the high-speed railway between London and Paris.

Arup is an intensive user of information technology in all aspects of its work, including working with clients, designing buildings, running structural simulations, and coordinating projects. Its management wants to ensure that Arup's information systems group is working on all the right IT projects for furthering the business and is doing so in the right way. Arup's systems have to be stable, leading edge, and available at all times, with employees able to access the information they need at any time and any place.

Until recently Arup's IT staff relied on Microsoft Excel spreadsheets or Microsoft Word documents as their project management tools. Reports were sporadic and in diverse formats, collaboration was very limited, project delivery styles were inconsistent, and there was no central visibility into what was happening with each project. Arup set up a Global IT Portfolio Management Office to oversee its entire portfolio of IT projects, but it was hampered by having to manually create reports using spreadsheets and email updates from regional offices.

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Working with Program Framework consultants who specialize in project portfolio management, Arup decided to adopt Microsoft Project Online to improve project management. Project Online is Microsoft's cloud-based project management tool, and it helps organizations efficiently plan projects, track status, and collaborate with others from any location and any device. Members of Arup's global workforce have immediate access to project data at any time wherever they are working. The cloud solution also makes it possible to report on projects using live data, with the system able to tie in to other processes such as service and change management. Program Framework consultants helped Arup implement Project Online and train employees. They also developed a customized Project and Program Status Reporting capability for Project Online.

In the past, Arup's Global IT Portfolio Management Office had to spend 40 hours per month compiling reports manually. By the time it created a status report, the report was already out of date. Project Online gives Arup instant views into the status of all of its IT projects. Regional employees can view their own portfolios of projects, while Arup's Global IT Portfolio Management Office has immediate views of all global projects. Arup's management can examine and classify projects throughout the entire enterprise based on their red, green, and amber status indicators. (Red designates projects with critical status, while amber designates those at risk.) The ability to see Arup's entire project portfolio gives management better insight into project delivery. The Global IT Portfolio Management Office can obtain key project status summaries, and highlight reporting of individual projects where it can drill down for further detail, enabling it to make better decisions based on up-to-date data. Project Online has become essential for supporting a common approach to Arup's project management across the globe. There is less duplication of effort and more strategic value in Arup's overall project portfolio.

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Project Online is part of Microsoft's cloud-based Office 365 software suite, so it works seamlessly with other Microsoft productivity and communication tools such as OneDrive for Business (cloud storage), Skype for Business (voice, video, chat), Yammer (enterprise social networking), and Visual Studio Team Foundation Server, which Arup uses for software development projects. Arup also plans to implement additional Project Online capabilities for demand and capacity planning, portfolio prioritization, and portfolio balancing. Users can easily copy information from Project and paste it into Office applications like PowerPoint and Word.

Arup uses Project Online for its IT Project Pipeline, a central repository of ideas for future development. Each idea recorded in the Pipeline requires that the initiator furnish information such as project description, budget, and resource needs. Arup's Global IT Portfolio Management Office sends this information to Arup's management committee members to review and prioritize for new initiatives.

When ideas are approved, their Project Pipeline information can easily be transferred to active projects. It only takes a few minutes for Project Pipeline to create a project or program within Project Online. Each has its own Project Details Pages, which include a built-in schedule template and a connected Microsoft SharePoint Server site with document repository and status reporting. This capability saves Arup's Global IT Portfolio Management Office manager Carolyn Bundey several days of work for each new project, creating significant time savings for an annual portfolio of approximately 180 IT projects.

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Several years ago, Project Online had about 150 users, but Arup is thinking about providing the tool for all of its employees. Arup licenses three different versions of Project Online. Project managers, owners, and administrators use Project Online with Project Professional for Office 365, enabling them to create and edit project plans inside or outside a web browser. Arup executives use Project Online to review project status. Project team members can view assignments or collaborate with other team members using the lower-cost Project Lite version.

Sources: "Engineering Firm Uses Cloud-Based Solution to Generate, Execute, and Monitor IT Projects," www.microsoft.com, accessed January 2, 2018; "Leading Arup at the Forefront of Innovation in Today's Built Environment," www.gineersnow.com, accessed January 3, 2018; and www.arup.com, accessed May 2, 2020.

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Arup Moves Project Management to the Cloud

- ***Week 12 Homework Questions***
  1. What is the relationship between information technology, project management, and Arup's business model and business strategy?
  2. How does Microsoft Project Online support Arup's business strategy? How did it change the way the company works?
  3. What management, organization, and technology issues did Arup have to address when selecting Project Online as its global project portfolio management tool?

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